

Research Paper

Publications of the Astronomical Society of Australia: L^AT_EX Guidelines for authors

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Abstract

This guide is for authors who are preparing papers for the *Publications of the Astronomical Society of Australia* journal using the L^AT_EX document preparation system and the CUP PAS style file.

Keywords: Key1, Key2, Key3, Key4

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1. Introduction

The layout design for the *Publications of the Astronomical Society of Australia* journal has been implemented as a L^AT_EX style file. The PAS style file is based on the ARTICLE style as discussed in the L^AT_EX manual. Commands which differ from the standard L^AT_EX interface, or which are provided in addition to the standard interface, are explained in this guide. This guide is not a substitute for the L^AT_EX manual itself.

1.1. Introduction to L^AT_EX

The L^AT_EX document preparation system is a special version of the T_EX typesetting program. L^AT_EX adds to T_EX a collection of commands which simplify typesetting by allowing the author to concentrate on the logical structure of the document rather than its visual layout.

L^AT_EX provides a consistent and comprehensive document preparation interface. There are simple-to-use commands for generating a table of contents, lists of figures and/or tables, and indexes. L^AT_EX can automatically number list entries, equations, figures, tables, and footnotes, as well as parts, chapters, sections and subsections. Using this numbering system, bibliographic citations, page references and cross references to any other numbered entity (e.g. chapter, section, equation, figure, list entry) are quite straightforward.

1.2. The PAS document class

The use of document class allows a simple change of style (or style option) to transform the appearance of your document. The CUP PAS class file preserves the standard L^AT_EX interface such that any document which can be produced using the standard L^AT_EX

ARTICLE style can also be produced with the PAS style. However, the fonts (sizes) and measure of text is slightly different from that for ARTICLE, therefore line breaks will change and it is possible that equations may need re-setting.

2. Additional facilities

In addition to all the standard L^AT_EX design elements, the PAS style includes the following feature:

- Extended commands for specifying a short version of the title and author(s) for the running headlines.

Once you have used this additional facility in your document, do not process it with a standard L^AT_EX style file.

2.1. Titles authors' names and affiliation

In the PAS style, the title of the article and the author's name (or authors' names) are used both at the beginning of the article for the main title and throughout the article as running headlines at the top of every page. The title is used on odd-numbered pages (rectos) and the author's name appears on even-numbered pages (versos). Although the main heading can run to several lines of text, the running head line must be a single line.

Moreover, the main heading can also incorporate new line commands (e.g. `\`) but these are not acceptable in a running headline. To enable you to specify an alternative short title and author's name, the standard `\lefttitle` commands have been used to print the running headline. If more authors has to be used in `\author` command then each authors should be captured in separate `\author` command. `\affiliation` command is used to call the affiliation, if more affiliations has to be used in `\affiliation` command then each affiliations should be captured in separate `\affiliation` command. `\email` command should be used inside the affiliation as shown below.

`\lefttitle{Cambridge Author}`

Author for correspondence: A. Cambridge, Email: ACambridge@student.unimelb.edu.au

Cite this article: Author1 C and Author2 C, an open-source python tool for simulations of source recovery and completeness in galaxy surveys. *Publications of the Astronomical Society of Australia* **00**, 1–12. <https://doi.org/10.1017/pasa.xxxx.xx>

```
\LaTeX\ Guidelines for~authors}
\title{The full title which can be as
long as necessary}
\author{Author's name}
\affiliation{the affiliation if necessary}
```

2.2. Abstract

The PAS style provides for an abstract which is produced by the following commands

```
\begin{abstract} ... \end{abstract}
```

2.3. Keyword

The PAS style provides for an Keyword which is produced by the following commands

```
\begin{keywords} ... \end{keywords}
```

2.4. Lists

The PAS style provides the three standard list environments.

- Bulleted lists, created using the `itemize` environment.
- Numbered lists, created using the `enumerate` environment.
- Labelled lists, created using the `description` environment.

2.5. Footnotes

The PAS journal style uses superior numbers for footnote references.^a

3. Some guidelines for using standard facilities

The following notes may help you achieve the best effects with the PAS style file.^b

3.1. Sections

$\LaTeX$ provides five levels of section headings and they are all defined in the PAS style file:

- `\section.`
- `\subsection.`
- `\subsubsection.`
- `\paragraph.`
- `\subparagraph.`

Section numbers are given for sections, subsection and subsubsection headings.

3.2. Running headlines

As described above, the title of the article and the author's name (or authors' names) are used as running headlines at the top of

every page. The title is used on odd-numbered pages (rectos) and the author's name appears on even-numbered pages (versos).

3.3. Tables

The `figure` and `table` environments are implemented as described in the $\LaTeX$ Manual to provide consecutively numbered floating inserts for illustrations and tables respectively. The standard inserts and their captions are formatted centred. Line breaks in captions can be inserted as required using `\\`.

The PAS style file will cope with most positioning of your tables and you should not normally use the optional positional qualifiers on the `table` environment which would override these decisions. Normal journal style sets the table caption first, followed by a double rule, the table body and a double rule at the bottom. Single rules and spanner rules (`\cline`) can be used to separate headings from the columns. For example, Table 1 is produced using the following commands:

```
\begin{table}
\caption{Results of Overloading for 3
Experimental Setups}\label{sample-table}
\tablefont\begin{tabular}{@{\extracolsep{\fill}}lcrrrrr}
\toprule
Program& Expt.&
CPU$\sim\{\rm a\}$$\& RelCPU$\sim\{\rm b\}$$\&
GC& Mem\footnote{Bytes of heap used over
the duration of the program.}&
RelMem\footnote{Memory usage relative
to experient (a).}\&
\hline
8 Queens& (a)& 2 88& 1 00& 6& 1 7M& 1 00\\
& (b)& 32 51& 11 29& 193& 48 9M& 28 76\\
\multirow{4}{*}{[3pt]{Primes}}& (a)& 4 89
& 1 00& 19& 5 3M& 1 00\\
& (b)& 47 54& 9 72& 204& 54 5M& 10 28\\
& (c)& 10 08& 2 06& 47& 13 0M& 2 45
\botrule
\end{tabular}}
\begin{tabnote}
{\$~{\rm a})$Input redshift of the simulated galaxy.}\tnp
{\$~{\rm b})$Magnitude bin that represents the
median value of the bins.}\tnp
\end{tabnote}
\end{table}
```

Notice the use of the `macro` to obtain the centered decimal points, inside the body of the table.

The `tabular` environment should be used to produce ruled tables; it has been modified for the PAS style in the following ways:

1. Additional vertical space is inserted above and below a horizontal rule (produced by `\hline`);

Table 1. Results of Overloading for 3 Experimental Setups

Program	Expt.	CPU ^a	RelCPU ^b	GC	Mem	RelMem
8 Queens	(a)	2 88	1 00	6	1 7M	1 00
	(b)	32 51	11 29	193	48 9M	28 76
	(a)	4 89	1 00	19	5 3M	1 00
Primes	(b)	47 54	9 72	204	54 5M	10 28
	(c)	10 08	2 06	47	13 0M	2 45

^aThis shows how a footnote is typeset.

^bTo know more information about $\LaTeX$ and its packages, try <https://ctan.org/?lang=en>

^aInput redshift of the simulated galaxy.

^bMagnitude bin that represents the median value of the bins.

Figure 1. An example figure with space for artwork.

2. Tables are centred, and span the full width of the page; that is, they are similar to the tables that would be produced by `\begin{minipage}{\textwidth}`.

Because of this reformatting, vertical rules should not be used; furthermore, commands to redefine quantities such as `\arraystretch` should be omitted. If the old tabular facilities are needed, there is a new environment, `oldtabular`, which has none of the reformatting; it should be used in exactly the same way.

3.4. Illustrations (or figures)

The PAS style will cope with most positioning of your illustrations and you should not normally use the optional positional qualifiers on the figure environment which would override these decisions. Figure captions should be below the figure itself, therefore the `\caption` command should appear after the figure or space left for an illustration.

Figure 1 shows an example onw working with LaTeX code to load art files. `\includegraphics` commnad is to load art files scale option used in `\includegraphics` is to reduce the art. EPS format will be compiled using LaTeX. PNG, PDF and JPG format art files are loaded in the same command but the TeX file should be compiled using PDFLaTeX:

```
\begin{figure}
\includegraphics[scale=.4]{sample.eps}
\caption{An example figure with space for artwork.}
\label{sample-figure}
\end{figure}
```

The vertical depth should correspond roughly to the artwork you will submit; it will be adjusted to fit the final artwork exactly.

3.5. Creating new theorem-like environments

You can create your own environments in $\text{\LaTeX}$, and although you may already be familiar with `\newtheorem`, you will not have seen the other two commands explained below.

`\newtheorem` is a standard command used for creating new theorem-like environments, such as theorems, corollaries, lemmas, conjectures and propositions, with the body of the text (automatically) in italic.

4. Mathematics

The PAS class file will centre displayed mathematics, and will insert the correct space above and below if standard $\text{\LaTeX}$ commands are used; for example use `\[ ... \]` and *not* `$$ ... $$`. Do not leave blank lines above and below displayed equations unless a new paragraph is really intended.

`amsmath.sty` is common package to handle various type math equations. The `amsmath` descriptions are available in the document can be find in the web link <https://ctan.org/pkg/amsmath?lang=en>

4.1. Numbering of equations

The `subequations` and `subeqnarray` environments have been incorporated into the PAS class file (see Section 4.1.1 regarding the `subequations` environment). Using these two environments, you can number your equations (1a), (1b) etc. automatically. For example, you can typeset

$$a_1 \equiv (2\Omega M^2/x)^{\frac{1}{4}} y^{\frac{1}{2}} \quad (1a)$$

and

$$a_2 \equiv (x/2\Omega)^{\frac{1}{2}} k_y/M. \quad (1b)$$

by using the `subequations` environment as follows:

```
\begin{subequations}
\begin{equation}
a_1 \equiv (2\Omega M^2/x)^{\textstyle\frac{1}{4}} y^{\textstyle\frac{1}{2}} \label{a1}
\end{equation}
and
\begin{equation}
a_2 \equiv (x/2\Omega)^{\textstyle\frac{1}{2}} k_y/M. \label{a2}
\end{equation}
\end{subequations}
```

4.1.1. The `subequations` environment and the `AMSTEX` package

The `amstex` (and the `amsmath`) packages also define a `subequations` environment. The environment in `PAS.cls` is used by default, as the environments in the AMS packages don't produce the correct style of output.

Note that the `subequations` environment from the `amstex` package takes an argument – you should use an 'a' to give α style `subequations`. e.g.

```
\begin{subequations}{a} ... \end{subequations}
```

4.2. Bibliography

As with standard $\text{\LaTeX}$, there are two ways of producing a bibliography; either by compiling a list of references by hand (using a `thebibliography` environment), or by using BibTeX with a suitable bibliographic database with the `bibliography` style provided with the `pasguide.tex` like `\bibliographystyle{paslike}`. The `pas.bst` will produce the bibliography which is similar to PAS style but not exactly. If any modification has to be made with `pas.bst` can be adjusted during manuscript preparation but the updated `bst` file should be given with source files. However, contributors are encouraged to format their list of references style outlined in section 4.2.2 below.

4.2.1. References in the text

References in the text are given by author and date. Whichever method is used to produce the bibliography, the references in the

text are done in the same way. Each bibliographical entry has a key, which is assigned by the author and used to refer to that entry in the text. There is one form of citation – `\cite{key}` – to produce the author and date. Thus, Salton *et al.* (1990) is produced by

```
\cite{sal90}.
```

`natbib.sty` is common package to handle various reference and its cross citations. The `natbib` descriptions are available in the document can be find in the web link <https://ctan.org/pkg/natbib?lang=en>

4.2.2. List of references

The following listing shows some references prepared in the style of the journal.

```
\bibitem[\protect\citenamename{Akmajian and Lehrer, }1976]{akm76}
Akmajian & Lehrer A. 1976, NP-like quantifiers and the
problem of determining the head of an NP. {\it Linguistic
Analysis\} 11, 295--313.
\bibitem[\protect\citenamename{Huddleston, }1984]{hud84}
Huddleston, Rodney. 1984, {\it Introduction to the Grammar of
English}. Cambridge: Cambridge University Press.
\bibitem[\protect\citenamename{McCord, }1990]{mcc90}
McCord, Michael C. 1990, Slot grammar: a system for simpler
construction of practical natural language grammars. In R.
Studer (ed.), {\it Natural Language and Logic: International
Scientific Symposium}, pp.~118--45. Lecture Notes in Computer
Science. Berlin: Springer-Verlag.
\bibitem[\protect\citenamename{Salton {\it et al.}, }1990]{sal90}
Salton, Gerald, Zhao, Zhongnan & Buckley, Chris. 1990,
A simple syntactic approach for the generation of indexing
phrases. Technical Report 90--1137, Department of Computer
Science, Cornell University.
\end{thebibliography}
```

This list typesets as shown at the end of this guide. Each entry takes the form

```
\bibitem[\protect\citenamename{Author(s), }Date]{tag}
Bibliography entry
```

where `Author(s)` should be the author names as they are cited in the text, `Date` is the date to be cited in the text, and `tag` is the tag that is to be used as an argument for the `\cite{}` command. `Bibliography entry` should be the material that is to appear in the bibliography, suitably formatted. This rather unwieldy scheme makes up for the lack of an author-date system in $\LaTeX$.

4.3. Catchline and date commands

To be placed in the preamble; for example:

- `\jnlDoiYr{2019}`
- `\doival{10.1017/xxxxx}`
- `\jnlPage{1}{4}`

4.4. Editing citations (when the author has used the `\cite` command)

In the past when an automatic `\cite` command produced text in the output which needed to be changed, the argument (in `[ ]`) from the bibliography entry was copied to the location of the `\cite`

command and then modified. The `\cite` command would then be removed as part of this process.

In the near future, we will probably have to supply $\TeX$ output which will need to contain ‘PDF marks’ for interactive browsing. Clearly by removing the automatic link to the bibliographic entry (referenced by the `\cite`), we are making extra work for ourselves later on.

To avoid this, the function of the `\cite` command’s optional argument has been changed. For example, the `\cite` command for the ‘mcc90’ entry gives:

McCord (1990)

but you want the following to appear in the text:

McCord (1990, see p. 119)

you would then use:

```
\cite[McCord (1990), see p.~119]{mcc90}
```

to obtain the desired result. Notice that you have to supply the round brackets as well in the optional argument.

Data Availability

We encourage authors to include a Data Availability Statement in their manuscript. This statement should include information on where resources such as data, materials, protocols and software code can be accessed. If data sharing is not applicable, authors should state that ‘Data sharing is not applicable to this article as no new data were created or analysed in this study.’

References

- Akmajian & Lehrer A. 1976, NP-like quantifiers and the problem of determining the head of an NP. *Linguistic Analysis* 11, 295–313.
- Huddleston, Rodney. 1984, *Introduction to the Grammar of English*. Cambridge: Cambridge University Press.
- McCord, Michael C. 1990, Slot grammar: a system for simpler construction of practical natural language grammars. In R. Studer (ed.), *Natural Language and Logic: International Scientific Symposium*, pp. 118–45. Lecture Notes in Computer Science. Berlin: Springer-Verlag.
- Salton, Gerald, Zhao, Zhongnan & Buckley, Chris. 1990, A simple syntactic approach for the generation of indexing phrases. Technical Report 90–1137, Department of Computer Science, Cornell University.